

Anime Face Alignment via Guideline-Based Transform for Expression Analysis

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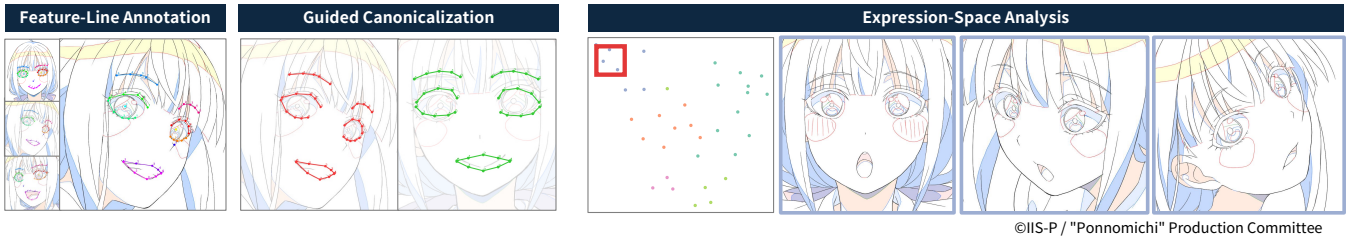


Figure 1: Overview of our framework for anime-style expression analysis. Sparse feature-line annotations guide canonicalization into a frontal pose, enabling pose-invariant expression clustering across different face orientations.

Abstract

We present a lightweight and interpretable framework for analyzing anime-style facial expressions using sparse, semantically meaningful feature-line annotations. By deforming annotated contours such as eyelids, eyebrows, and lips into a canonical frontal pose, our method removes pose variation while preserving expressive differences. The canonicalized space enables consistent expression clustering and transfer across views, providing a practical foundation for stylized face normalization in anime production.

CCS Concepts

• Computing methodologies → Non-photorealistic rendering.

Keywords

Anime Face Analysis, Feature-Line Annotation, Canonicalization, Expression Analysis, Stylized Character Processing

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1 Introduction

Analyzing anime-style facial expressions is important for character design and expressive animation. Unlike photorealistic faces, anime faces exhibit artistic exaggerations and stylized deformations that invalidate standard alignment techniques based on 3D geometry or dense landmarks.

We propose a lightweight and interpretable framework using sparse, semantically meaningful feature-line annotations. Key contours such as eyelids, eyebrows, and lips guide a structure-preserving deformation into a canonical frontal pose, removing pose-dependent variation while preserving expressive differences. The canonicalized representations enable consistent expression clustering and part-wise analysis inspired by the Facial Action Coding System (FACS), as well as expression transfer across views.

Conventional alignment methods [Banz and Vetter 1999; Deng et al. 2019] assume photorealistic geometry, while recent generative and sketch-based techniques [Huang et al. 2023; Li et al. 2021; Simo-Serra et al. 2016] lack structural interpretability. In contrast, our approach builds on feature-line guidance [Beier and Neely 1992] to achieve pose-invariant normalization suitable for stylized anime faces.

Our contributions are threefold:

- A feature-line-guided canonicalization framework robust to stylization and pose variation.
- A sparse annotation scheme enabling both global and part-wise expression analysis.
- Demonstrations of expression-space clustering and transfer for production-oriented applications.

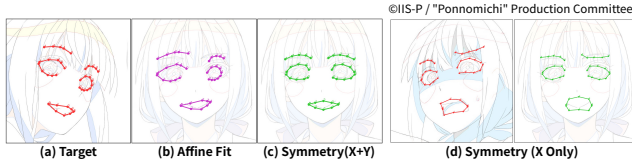


Figure 2: Guided canonicalization of feature lines: (a) target, (b) affine fit correcting scale and rotation, (c) symmetry correction yielding a canonical frontal view, and (d) x-only symmetry correction variant.

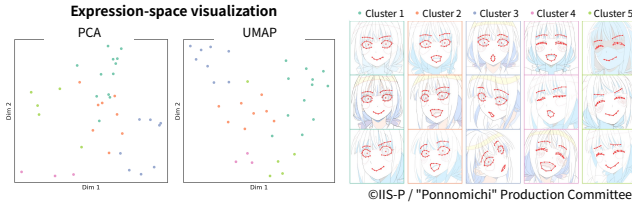


Figure 3: Clustering results across all layers. Left: 2D embedding (PCA/UMAP) showing consistent expression clusters across face orientations. Right: representative samples from five clusters, each illustrating characteristic expression patterns.

2 Guided Canonicalization and Expression Analysis

We present a lightweight framework for pose-invariant analysis of anime-style facial expressions using sparse, semantically meaningful feature-line annotations. Key contours such as eyelids, eyebrows, and lips are manually annotated and used to guide a structure-preserving deformation into a canonical frontal pose. This guided canonicalization removes pose-dependent distortion while preserving expressive variation, enabling consistent comparison across views.

Canonicalization. As shown in Figure 2, we first estimate an affine transformation to correct global scale and rotation, followed by a symmetry correction along the facial midline. This yields a canonical frontal representation suitable for consistent expression analysis.

Expression Analysis. We concatenate canonicalized feature-line coordinates and visualize the expression distribution using PCA or UMAP embedding. As shown in Figure 3, similar expressions form consistent clusters even when the original faces differ in orientation. Part-wise clustering (Figure 4) further decomposes local motions such as eyebrow raising or eyelid closure, resembling FACS-style analysis.

Expression Transfer. The canonicalized representation also enables smooth expression transfer across views. By interpolating deformation between canonical and target spaces, expressions can be consistently applied to different views while preserving stylistic coherence. This highlights the potential for blendshape design and expression-aware production workflows.

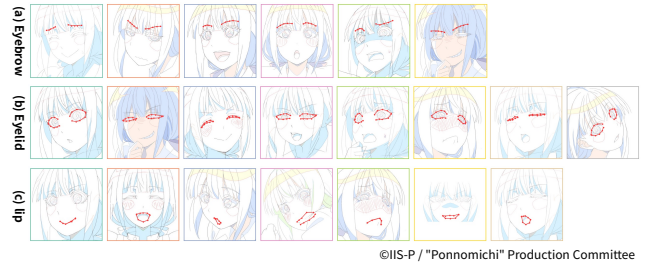


Figure 4: Part-wise clustering results: (a) eyebrow (6), (b) eyelid (8), and (c) lip (7) clusters. Each shows one representative sample illustrating characteristic expression patterns.

3 Discussion and Conclusion

Our results demonstrate that sparse, semantically meaningful feature-line annotations enable pose-invariant analysis of anime-style facial expressions. The canonicalized representation supports FACS-like clustering and part-wise decomposition, providing a practical foundation for dataset organization, blendshape design, and expression transfer in production.

Key applications include:

- Structuring expression datasets and selecting representative templates.
- Designing consistent blendshapes for animation production.
- Transferring expressions across views via feature-line mapping.

Future work includes automating annotation, extending to 3D and multi-character setups, and integrating FACS-style slider control for interactive editing. We believe our framework offers a lightweight and interpretable basis for stylized face normalization and expression-aware animation workflows.

Acknowledgments

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References

- Thaddeus W Beier and Shawn Neely. 1992. Feature-based image metamorphosis. *ACM SIGGRAPH Computer Graphics* 26, 2 (1992), 35–42.
- Volker Blanz and Thomas Vetter. 1999. A morphable model for the synthesis of 3D faces. *Proceedings of the 26th annual conference on Computer graphics and interactive techniques* (1999), 187–194.
- Yu Deng, Jiankang Feng, Jia Hu, et al. 2019. Accurate 3D face reconstruction with weakly-supervised learning: From single image to image set. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 6013–6022.
- Zhengyu Huang, Haoran Xie, Tsukasa Fukusato, and Kazunori Miyata. 2023. Aniface-drawing: Anime portrait exploration during your sketching. In *ACM SIGGRAPH 2023 conference proceedings*. 1–11.
- Bing Li, Yuanlue Zhu, Yitong Wang, Chia-Wen Lin, Bernard Ghanem, and Linlin Shen. 2021. Anigan: Style-guided generative adversarial networks for unsupervised anime face generation. *IEEE Transactions on Multimedia* 24 (2021), 4077–4091.
- Edgar Simo-Serra, Satoshi Iizuka, Kazuma Sasaki, and Hiroshi Ishikawa. 2016. Learning to simplify: Fully convolutional networks for rough sketch cleanup. *ACM Transactions on Graphics (TOG)* 35, 4 (2016), 1–11.